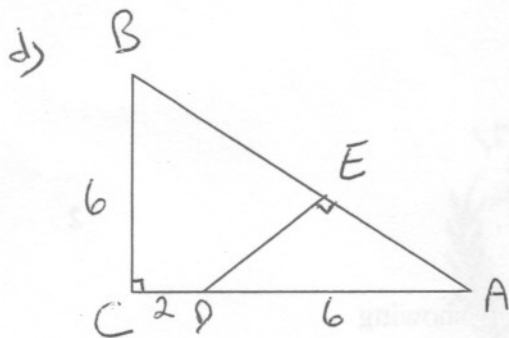


Question One

- a) Solve $6x^2 - 5x + 1 = 0$ 2
- b) Find the value of y if $a^{4y-3} = \frac{1}{\sqrt{a}}$ 2
- c) If $\cos A = \frac{2}{5}$ and $\tan A < 0$ find the exact value of $\sin A$ and $\cot A$. 3
- d) i) P divides the interval from A(-4,-6) to B(6,-1) in the ratio k:2. Write down the coordinates of P. 2
- ii) If P lies on the line $4x - 3y = 17$ find the exact coordinates of P. 2
- e) i) Neatly sketch $y = |3x - 6|$ 2
- ii) Hence or otherwise solve $|3x - 6| = 2x - 5$ 2

Question Two (Start a new page)

- a) Given that $f(x) = x^2 + x$, simplify $\frac{f(x-1)}{x-1}$ 2
- b) If the perpendicular distance between (-1, 3) and the line $ax + 3y - 15 = 0$ is 2 units, find the exact value/s of a . 2
- c) i) Is the function $y = \frac{2}{x^2}$ odd or even? Justify your answer. 2
- ii) Neatly sketch $y = \frac{2}{x^2}$, clearly showing all important features. 2



In $\triangle ABC$, $AD = 6$ units, $DC = 2$ units and $BC = 6$ units.

i) Prove $\triangle ABC \parallel \triangle ADE$.

3

ii) Find DE , giving reasons.

2

e) Simplify $\frac{1}{2} [\sin(A + B) - \sin(A - B)]$

2

Question Three (Start a new page)

a) Differentiate the following with respect to x :

i) $3x^3 - x^5$

1

ii) $x\sqrt{x}$

2

iii) $(x^2 + 1)^2$

2

iv) $\frac{\ln x}{x}$

2

b) When is $f(x) = 2x^3 - \frac{5}{2}x^2 + x + 6$ concave up?

2

c) Find the exact gradient of the tangent to $y = \sin x \cdot \tan x$ when $x = \frac{\pi}{6}$

3

d) Find $\lim_{x \rightarrow 2} \frac{x-2}{x+3}$

1

e) If the roots of $kx^2 - 2x + (1 - 2k) = 0$ are reciprocals, find the exact value of their sum.

2

Question Four (Start a new page)

- a) Write down the domain and range of $y = \sqrt{x^2 - 4}$. 2
- b) Neatly sketch using calculus $y = 2x^3 - \frac{5}{2}x^2 + x + 6$ showing all important features. 4
- c) Find the locus of a point $P(x, y)$ which moves so that its distance from $A(a, 2a)$ is equal to its distance from $B(2a, a)$. 2
- d) Find the values of k for which $y = e^{-kx}(x - k)$ is stationary at $x = 2.5$ 3
- e)



- i) Prove that $\cos x = \frac{b}{2a}$ (Hint use the sine rule). 2
- ii) Prove that $c = \frac{b^2 - a^2}{a}$ or $c = a$ 2

END OF PAPER

Question One

a) $6x^2 - 5x + 1 = 0$
 $\times 6, -5$

$6x^2 - 3x - 2x + 1 = 0$

$3x(2x-1) - (2x-1) = 0$

$(3x-1)(2x-1) = 0$

$\therefore x = \frac{1}{3} \text{ or } x = \frac{1}{2}$

(2)

b) $a^{4y-3} = \frac{1}{\sqrt{a}}$

$a^{4y-3} = a^{-\frac{1}{2}}$

$\therefore 4y-3 = -\frac{1}{2}$

$4y = 2\frac{1}{2}$

$\therefore y = \frac{5}{8}$

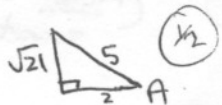
(1)

(1)

(or $y = 0.625$)

c) $\cos A = \frac{2}{5}$

$\frac{\text{adj}}{\text{hyp}} = \frac{2}{5}$



$\therefore \sin A = \frac{4}{5}$

$\cos A = -\frac{2}{\sqrt{21}}$

d) i) $(-4, -6) \quad (6, -1)$

$k: 2$

$\therefore P = \left(\frac{-8+6K}{K+2}, \frac{-12-K}{K+2} \right)$

(1)

(1)

ii) $4 \frac{(-8+6K)}{K+2} - 3 \frac{(-12-K)}{K+2} = 17$

$-32 + 24K + 36 + 3K = 17K + 34$

$24K - 17K = 34 - 4$

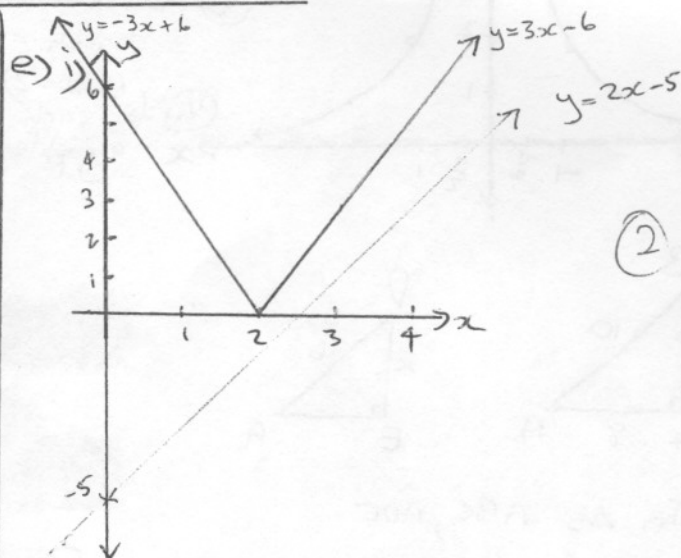
$10K = 30$

$\therefore K = 3$

(1)

$\therefore P$ is $(2, -3)$

(1)



(2)

ii) $|3x-6| = 2x-5$

no real solution (1)

(+ 1 mark for graph or 1 mark for solving algebraically and 1 mark for explaining that there's no real solution.)

Question Two

a) $\frac{f(x-1)}{x-1} = \frac{(x-1)^2 + (x-1)}{x-1}$

but $x \neq 1$

(1)

$= x-1+1$
 $= x$

(1)

b) $d = \frac{|ax+by+c|}{\sqrt{a^2+b^2}}$

$2 = \frac{|-a+9-15|}{\sqrt{a^2+9}}$

(1)

$4 = \frac{(-a-6)^2}{a^2+9}$

$4a^2+36 = a^2+12a+36$

$3a^2-12a = 0$

$3a(a-4) = 0$

$\therefore a = 0 \text{ or } 4$

(1)

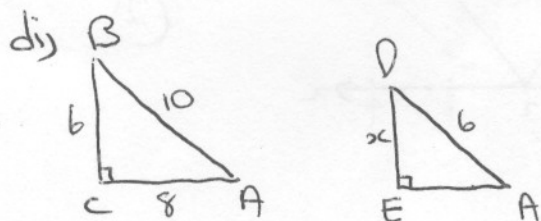
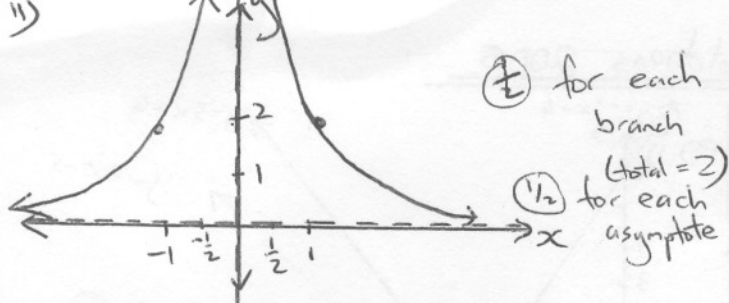
c) i) $f(x) = \frac{2}{x^2}$

$f(-x) = \frac{2}{(-x)^2} = \frac{2}{x^2} = f(x)$

(1)

$\therefore$ even function

(1)



In Δ s ABC, ADE

$\hat{A}$ is common

$\hat{BCA} = \hat{DEA}$ (both 90°)

$\therefore \Delta ABC \parallel \Delta ADE$ (equiangular)

ii) $\frac{x}{6} = \frac{6}{10}$ (corresponding sides in similar triangles are in the same ratio)

$x = 3\frac{3}{5}$ units

e) $\frac{1}{2} [\sin(A+B) - \sin(A-B)]$

$= \frac{1}{2} [\sin A \cos B + \cos A \sin B - \sin A \cos B + \cos A \sin B]$

$= \frac{1}{2} \times 2 \cos A \sin B$

$= \cos A \sin B$

Question Three

i) $9x^2 - 5x^4$

ii) $y = x^{3/2}$

$\therefore y' = \frac{3}{2} x^{1/2}$

$= \frac{3}{2} \sqrt{x}$

iii) $y = x^4 + 2x^2 + 1$

$\frac{dy}{dx} = 4x^3 + 4x$

iv) $y = \frac{\ln x}{x}$

$y' = \frac{x \cdot \frac{1}{x} - \ln x}{x^2} = \frac{1 - \ln x}{x^2}$

b) $f(x) = 2x^3 - \frac{5}{2}x^2 + x + 6$

$f'(x) = 6x^2 - 5x + 1$

$f'(x) = 12x - 5$

concave up when

$f''(x) > 0$

$\therefore 12x - 5 > 0$

$12x > 5$

$\therefore x > \frac{5}{12}$

c) $y = \sin x \cdot \tan x$

$\frac{dy}{dx} = \tan x \cdot \cos x + \sin x \cdot \sec^2 x$

$= \sin x + \tan x \cdot \sec x$

when $x = \frac{\pi}{6}$,

$m = \sin \frac{\pi}{6} + \tan \frac{\pi}{6} \cdot \frac{1}{\cos \frac{\pi}{6}}$

$= \frac{1}{2} + \frac{1}{\sqrt{3}} \times \frac{2}{\sqrt{3}}$

$= 1\frac{1}{6}$ (must be exact!!)

d) $\lim_{x \rightarrow 2} \frac{x-2}{x+3}$

$= \frac{0}{5}$

$= 0$

e) let roots be α and $\frac{1}{\alpha}$

$\therefore \alpha + \frac{1}{\alpha} = \frac{2}{K}$

$1 = \frac{1-2K}{K}$

from (2) $K = 1 - 2K$

$3K = 1$

$K = \frac{1}{3}$

sub into (1)

$\therefore \alpha + \frac{1}{\alpha} = \frac{2}{1/3}$

$= 6$

Question Four

a) $y = \sqrt{x^2 - 4}$

D: $x \leq -2$ or $x \geq 2$ (1)

R: $y \geq 0$ (1)

b) $y = 2x^3 - \frac{5}{2}x^2 + x + 6$

$\frac{dy}{dx} = 6x^2 - 5x + 1$

stationary points when $\frac{dy}{dx} = 0$

ie at $x = \frac{1}{3}$ or $\frac{1}{2}$ (from 1a) (1)

$\frac{d^2y}{dx^2} = 12x - 5$

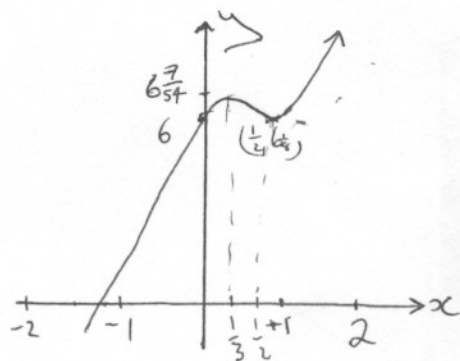
when $x = \frac{1}{3}$, $\frac{d^2y}{dx^2} = -1$, concave down

$\therefore$ max at $(\frac{1}{3}, \frac{67}{54})$ (1)

when $x = \frac{1}{2}$, $\frac{d^2y}{dx^2} = 1$, concave up

$\therefore$ min at $(\frac{1}{2}, \frac{1}{8})$ (1)

intercept (0, 6)



c) $PA = 2PB$

$(x-a)^2 + (y-2a)^2 = (x-2a)^2 + (y-a)^2$ (1)

$x^2 - 2ax + a^2 + y^2 - 4ay + 4a^2 = x^2 - 4ax + 4a^2 + y^2 - 2ay + a^2$

$2ax - 2ay = 0$ (1/2)

$2a(x-y) = 0$

$\therefore$ The locus is the line $y = x$. (1/2)

d) $y = e^{-kx}(x-k)$

$\frac{dy}{dx} = e^{-kx} + (-k)(x-k)e^{-kx}$

$= e^{-kx} [1 - k(x-k)]$

$= e^{-kx} (1 - kx + k^2)$

stat. points when $\frac{dy}{dx} = 0$ (1)

ie when $e^{-kx} (1 - kx + k^2) = 0$

but $e^{-kx} \neq 0 \therefore$ when $1 - kx + k^2 = 0$

now $x = 2.5$

$\therefore 1 - k(2.5) + k^2 = 0$

$k^2 - 2.5k + 1 = 0$

$k = \frac{2.5 \pm \sqrt{2.5^2 - 4}}{2}$

$= \frac{2.5 \pm 1.5}{2}$

$\therefore k = 2$ or $\frac{1}{2}$ (1)

e) i) $\frac{b}{\sin 2x} = \frac{a}{\sin x}$ (1)

$\frac{\sin 2x}{2 \sin x \cos x} = \frac{a}{b}$ (1/2)

$\frac{1}{\cos x} = \frac{2a}{b}$

$\therefore \cos x = \frac{b}{2a}$ (1/2)

(11)

PTO

an alternative solution for

(e) ii) using the cos rule

$$a^2 = b^2 + c^2 - 2bc \cos x$$

$$a^2 = b^2 + c^2 - 2bc \cdot \frac{b}{2a}$$

from (i)

$$a^2 = b^2 + c^2 - \frac{b^2 c}{a}$$

$$a^3 = ab^2 + ac^2 - b^2 c$$

$$\therefore ac^2 - b^2 c + ab^2 - a^3 = 0$$

$$\therefore c = \frac{b^2 \pm \sqrt{b^4 - 4a(ab^2 - a^3)}}{2a}$$

$$= \frac{b^2 \pm \sqrt{b^4 - 4a^2 b^2 + 4a^4}}{2a}$$

$$= \frac{b^2 \pm \sqrt{(b^2 - 2a^2)^2}}{2a}$$

$$= \frac{b^2 \pm (b^2 - 2a^2)}{2a}$$

$$= \frac{2b^2 - 2a^2}{2a} \quad \text{or} \quad \frac{2a^2}{2a}$$

$$= \frac{b^2 - a^2}{a} \quad \text{or} \quad a$$

